Q4 冷媒性能比较 / Refrigerant comparison

【1】逐图计算 / Calculations from the P-H diagrams

Refrigeration effect (q_e) = $h_1 - h_3$ (kJ/kg)

$$Coefficient\ \ of\ \ Performance\ \ (COP) = \frac{h_1 - h_3}{h_2 - h_1}$$

图 1 / Figure 1 — R-22

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 蓝色 / Blue	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	408	408.5
h_2 / kJ/kg	435	438
h_3 / kJ/kg	256	278
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	408 − 256 = 152	408.5 − 278 = 130.5
Compression work (w) h_2–h_1 / kJ/kg	435 − 408 = 27	438 − 408.5 = 29.5
Coefficient of Performance (COP)	$\frac{408 - 256}{435 - 408} = 5.63$	$\frac{408.5 - 278}{438 - 408.5} = 4.42$
Evaporating pressure p_e / bar(abs)	6	7
Condensing pressure p_c / bar(abs)	17	24.1
Specific volume (inlet) v_1 / m^3/kg	0.040	0.034
Specific volume (outlet) v_2 / m^3/kg	0.0155	0.0108
Entropy s / kJ/(kg·K)	1.75	1.73

图 2 / Figure 2 — R-407C

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 橙色 / Orange	Room Cooler: 10℃ / 60℃ 绿色 / Green
h_1 / kJ/kg	415	417
h_2 / kJ/kg	446	449
h_3 / kJ/kg	272	304
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	415 − 272 = 143	417 − 304 = 113
Compression work (w) h_2–h_1 / kJ/kg	446 − 415 = 31	449 − 417 = 32
Coefficient of Performance (COP)	$\frac{415 - 272}{446 - 415} = 4.61$	$\frac{417 - 304}{449 - 417} = 3.53$
Evaporating pressure p_e / MPa(abs)	0.55	0.64
Condensing pressure p_c / MPa(abs)	1.9	2.8
Specific volume (inlet) v_1 / m^3/kg	0.043	0.038
Specific volume (outlet) v_2 / m^3/kg	0.013	0.0082
Entropy s / kJ/(kg·K)	1.77	1.76

图 3 / Figure 3 — R-134a

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 绿色 / Green	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	401	404
h_2 / kJ/kg	426	434
h_3 / kJ/kg	263	287
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	401 − 263 = 138	404 − 287 = 117
Compression work (w) h_2–h_1 / kJ/kg	426 − 401 = 25	434 − 404 = 30
Coefficient of Performance (COP)	$\frac{401 - 263}{426 - 401} = 5.52$	$\frac{404 - 287}{434 - 404} = 3.90$
Evaporating pressure p_e / MPa(abs)	0.34	0.41
Condensing pressure p_c / MPa(abs)	1.2	1.7
Specific volume (inlet) v_1 / m^3/kg	0.060	0.053
Specific volume (outlet) v_2 / m^3/kg	0.017	0.014
Entropy s / kJ/(kg·K)	1.73	1.72

图 4 / Figure 4 — R-123

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 蓝色 / Blue	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	384	386
h_2 / kJ/kg	407	414
h_3 / kJ/kg	245	261
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	384 − 245 = 139	386 − 261 = 125
Compression work (w) h_2–h_1 / kJ/kg	407 − 384 = 23	414 − 386 = 28
Coefficient of Performance (COP)	$\frac{384 - 245}{407 - 384} = 6.04$	$\frac{386 - 261}{414 - 386} = 4.46$
Evaporating pressure p_e / MPa(abs)	0.04	0.051
Condensing pressure p_c / MPa(abs)	0.18	0.28
Specific volume (inlet) v_1 / m^3/kg	0.38	0.30
Specific volume (outlet) v_2 / m^3/kg	0.097	0.058
Entropy s / kJ/(kg·K)	1.67	1.66

【2】性能及构造比较 / Performance and construction comparison

各格数值按 Chiller（5℃/45℃）／Room Cooler（10℃/60℃）顺序列出；压力保留原图单位，1MPa=10bar。

Values are listed as Chiller (5℃/45℃) / Room Cooler (10℃/60℃). Original pressure units are retained; 1MPa = 10bar.

项目 / Item	R-22	R-123	R-134a	R-407C	比较及构造影响 Comparison and construction implications
Coefficient of Performance (COP) 制冷性能系数	5.63/4.42	6.04/4.46	5.52/3.90	4.61/3.53	两工况均 R-123＞R-22＞R-134a＞R-407C；同冷量下，COP 较高可减少压缩机功率及冷凝器散热量。 Both cases rank R-123 > R-22 > R-134a > R-407C. At equal duty, higher COP reduces compressor power and condenser heat rejection.
Refrigeration effect 制冷效应 q_e / kJ/kg	152/130.5	139/125	138/117	143/113	R-22 两工况均最大。同冷量下，较大 q_e 可降低质量流量；改为房间工况时，四者 q_e 及 COP 均下降、压缩功 w 均增加。 R-22 is highest in both cases. Larger q_e reduces mass flow at equal duty. Room-cooler conditions lower q_e/COP and increase w for all four.
Evaporating pressure 蒸发压力 p_e	6/7 bar(abs)	0.04/0.051 MPa(abs)	0.34/0.41 MPa(abs)	0.55/0.64 MPa(abs)	两工况均 R-22＞R-407C＞R-134a＞R-123。R-123 低于大气压，须良好密封、排不凝气体装置及水分控制。 Both rank R-22 > R-407C > R-134a > R-123. Sub-atmospheric R-123 needs good sealing, a purge unit and moisture control.
Condensing pressure 冷凝压力 p_c	17/24.1 bar(abs)	0.18/0.28 MPa(abs)	1.2/1.7 MPa(abs)	1.9/2.8 MPa(abs)	R-407C 最高、R-22 次高，设备须相应耐压及超压保护。R-407C 等压相变有温度滑移，须匹配换热器及 TEV。 R-407C is highest; R-22 follows. Provide suitable pressure ratings and protection. R-407C glide affects heat exchangers and Thermal Expansion Valve (TEV) control.
Specific volume 入口 v_1／出口 v_2 m^3/kg	v_1: 0.040/0.034 v_2: 0.0155/0.0108	v_1: 0.38/0.30 v_2: 0.097/0.058	v_1: 0.060/0.053 v_2: 0.017/0.014	v_1: 0.043/0.038 v_2: 0.013/0.0082	各组 $v_2＜v_1$；R-123 两处比容最大。同冷量下，吸气容积流量 $\dot{V}_1=\dot{Q}_e v_1/q_e$ 较大，利于采用 Centrifugal compressor；R-22/R-407C 较利于紧凑型机组。 All have $v_2 < v_1$; R-123 has the largest specific volumes. Its higher equal-duty suction flow, $\dot{V}_1 = \dot{Q}_e v_1/q_e$, favours a Centrifugal compressor. R-22/R-407C favour compact units.
Entropy 熵 s / kJ/(kg·K)	1.75/1.73	1.67/1.66	1.73/1.72	1.77/1.76	理想压缩取 $s_1=s_2$；实际绝热过程产熵、耗功增加、COP 下降。熵值不直接表示效率；R-123 须适当吸气过热以避免湿压缩。 Ideal compression uses $s_1 = s_2$. Real adiabatic compression generates entropy and needs more work. Entropy does not rank efficiency; R-123 needs suction superheat to avoid wet compression.

【3】安全比较 / Safety comparison

项目 / Item	R-22	R-123	R-134a	R-407C	比较及构造影响 Comparison and construction implications
Corrosion 腐蚀性	保持干燥 Keep dry	尤须防空气及水分吸入 Limit inward air/moisture leakage	保持干燥 Keep dry	保持干燥 Keep dry	四者均须相容的材料及润滑油，并保持系统干燥、洁净；水分、污染及高温分解物可引起腐蚀。 All four need compatible materials/lubricant and dry, clean systems. Moisture, contamination and thermal decomposition products can cause corrosion.
Explosion 爆炸及爆裂风险	1 级 / Class 1	1 级 / Class 1	1 级 / Class 1	1 级 / Class 1	1 级指标准试验下无火焰传播；高温、超压仍可造成设备爆裂。须有适当耐压、压力切断及泄压保护。 Class 1 means no flame propagation in classification tests. Heat/overpressure can still rupture equipment; provide suitable pressure ratings, cut-outs and relief protection.
Toxicity 毒性	A1	B1	A1	A1	R-123 毒性类别高于其余三者。四者均须控制泄漏并通风；R-123 尤其须注意人员暴露及泄漏检测，高浓度泄漏还可能造成缺氧。 R-123 has the higher toxicity class. Control leaks and ventilate all systems, with particular attention to R-123 exposure/detection. High concentrations can displace oxygen.

【4】环境影响 / Environmental impacts

Ozone Depletion Potential (ODP)｜臭氧层消耗潜能

R-22 及 R-123 为含氯 HCFC，课件 ODP 分别为 0.05 及 0.014；R-134a 及 R-407C 为 HFC，ODP 为 0。R-22/R-123 泄漏会破坏臭氧层，故须控制泄漏并回收冷媒。

Chlorine-containing HCFCs R-22 and R-123 have course-table ODPs of 0.05 and 0.014. HFCs R-134a and R-407C have zero ODP. R-22/R-123 leakage damages the ozone layer, requiring containment and refrigerant recovery.

Global Warming Potential (GWP)｜全球变暖潜能

统一采用 IPCC AR4 的 100 年 GWP：R-22 为 1810、R-123 为 77、R-134a 为 1430、R-407C 约 1774。四者泄漏均造成直接温室效应，R-123 较低；较高 COP 可减少耗电引致的间接排放。

On the same IPCC AR4 100-year basis, GWP is 1810 for R-22, 77 for R-123, 1430 for R-134a and about 1774 for R-407C. All cause direct warming when leaked, with R-123 lower. Higher COP reduces electricity-related indirect emissions.

资料 / Sources: 手绘循环数据及 DuPont P–H 图；HVAC System II, slides 9–20, 29, 47–48；ASHRAE Handbook—Fundamentals (2025), Ch.29；IPCC AR4 (2007), WGI Table 2.14, pp.212–213（GWP100；R-407C 按 R32/R125/R-134a 的 23/25/52 质量分数加权 / mass-weighted）。